

STEM Like A Girl

Breaking Barriers, Building Future

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ELI 345 - Grant Writing

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STEM Action Grant Committee
Society for Science
1719 N St NW
Washington, DC 20036-280

Dear STEM Action Grant Committee,

Since 2018, STEM Like a Girl has worked to inspire and equip girls with the confidence and curiosity to pursue STEM as future innovators and leaders. Our interactive workshops for 3rd–5th grade girls and their adult caregivers have fostered curiosity, strengthened family engagement, and reached more than 1,400 participants through 48 workshops, many from underrepresented and low-income communities.

We are requesting \$5,000 through the STEM Action Grant to expand our workshop offerings, develop new learning modules, and ensure continued financial assistance so that cost is never a barrier to participation. With your support, we can reach more families, deepen our community impact, and foster inclusive STEM learning environments where girls feel encouraged to explore and thrive.

With your support, we can continue building a welcoming space where girls and their caregivers learn, experiment, and discover the excitement of STEM together.

Sincerely,

Sarah Foster
Founder & Executive Director

SUMMARY

STEM Like a Girl, founded in 2018 in Portland, Oregon, empowers girls in grades 3–5 to build confidence, curiosity, and lasting engagement in STEM. Our mission is to excite and support girls as future problem solvers and leaders, ensuring that all girls, regardless of background or income, have access to joyful, inclusive STEM learning opportunities.

We deliver hands-on workshops where each girl participates alongside a trusted adult caregiver, strengthening family support, building confidence, and fostering belonging in scientific spaces. Since our founding, we have served more than 1,423 girls through 48 workshops across Oregon, Washington, and New Jersey. Over 41% of participants identify as minorities, and we offer full and partial financial assistance to ensure cost is never a barrier. Through partnerships with community organizations, we bring programs directly into underserved neighborhoods.

Our work addresses persistent gaps in access to hands-on STEM learning, encouragement, and representation, particularly for girls of color and girls from low-income families. By offering accessible, family-centered programming, we strengthen both STEM confidence and long-term engagement.

This proposal respectfully requests \$5,000 in STEM Action Grant funding to expand our community workshops and purchase new learning materials. Funding will support increased outreach to underserved families and continued scholarship support to ensure equitable access. By empowering girls and their caregivers to explore science together, this project advances the STEM Action Grant's mission to promote diversity, inclusion, and equity in STEM education and helps cultivate the next generation of innovators and leaders who reflect the diversity of our communities.

PROBLEM STATEMENT

The STEM Action Grant prioritizes programs that increase access, participation, and representation of underrepresented groups in STEM fields. It seeks to empower community-based initiatives that close opportunity gaps and encourage long-term interest in STEM careers.

Oregon continues to face persistent disparities in STEM participation among girls, especially those from low-income, rural, and BIPOC communities. According to the National Assessment of Educational Progress, Oregon ranks 47th nationally in overall education performance and even lower in science achievement (*National Data Explorer - ETS*, n.d.). In addition, U.S. Census data shows that women remain significantly underrepresented across STEM fields, making up only 28.2% of the STEM workforce. Research also indicates that many girls begin to lose interest in STEM as early as middle school due to limited confidence, insufficient exposure to hands-on learning, and a lack of mentors and role models (*STEM - Women in the States*, 2015).

STEM Like a Girl addresses this critical need by creating welcoming, hands-on learning experiences where girls and their caregivers can explore STEM together. Our workshops are designed to build confidence, curiosity, and a sense of belonging in STEM fields. However, funding limitations continue to restrict the number of workshops we can offer, limit our reach into new communities, and reduce our ability to provide full financial assistance for families who need it most.

Research indicates that early, positive exposure to STEM activities, especially when supported by family engagement, significantly increases girls' likelihood of pursuing advanced STEM learning (*UNESCO, 2023, pg 23-24*). Yet many families lack affordable, accessible opportunities for such experiences, particularly in underserved areas.

Without targeted intervention, these barriers will continue to widen the gender gap in STEM fields. The middle school years are a decisive turning point, research shows girls' interest in STEM drops by 30% between grades 5–8 if they lack encouragement and exposure (*Girl Scout Research Institute et al., 2012*).

If these disparities are not addressed, girls will continue to face barriers in education and the workforce, which limits diversity in STEM fields, reduces opportunities for innovation, and makes it harder for communities to meet future scientific and technological challenges.

OBJECTIVES

The primary goal is to expand access to high-quality, hands-on STEM learning for girls in grades 3–5 from underrepresented and low-income communities, increasing confidence, curiosity, and engagement in STEM.

Specific, Measurable Grant Project Activity Objectives

With support from the STEM Action Grant, STEM Like a Girl will:

- Host 2 extra community-based workshops, serving approximately 50–70 girls and their adult caregivers.
- Provide full or partial financial assistance to at least 40% of participating families, ensuring that cost is not a barrier to participation.
- Recruit and train 20–36 new volunteers, including women in STEM fields, to serve as mentors and facilitators.
- Collect and analyze participant data through pre- and post-workshop surveys to inform program improvements and measure outcomes.

Expected, Measurable Outcomes

The following outcomes will indicate project success:

- Participation Growth: At least a 25% increase in girls from underrepresented communities attending workshops.
- Confidence & Engagement: At least 85% of participants will report increased confidence and interest in STEM, which is an early indicator linked to higher persistence and graduation rates in STEM pathways.
- Family Engagement: 90% of caregivers will feel more confident supporting their child's STEM learning at home.
- Volunteer Impact: A 30% increase in active volunteers, strengthening program delivery and mentorship capacity.

Dollar Value of Grant Project Outcomes

A \$5,000 grant will directly support two STEM workshops and subsidize financial assistance for participating families. The investment creates impact well beyond its dollar amount:

- \$5,000 offsets the true program cost (approximately \$142 per participant) for 50–70 girls and their caregivers, ensuring equitable access.
- Workshop supplies and coordination support 8–10 hands-on STEM activities per session, resulting in an estimated 300–560 total hours of STEM learning across all participants.
- Because each workshop serves both girls and their caregivers, the investment strengthens family engagement in STEM, amplifying long-term learning outcomes without added cost.

PROJECT DESCRIPTION

Funding from the STEM Action Grant will directly support the expansion of STEM Like a Girl's evidence-based workshops. Each workshop runs 3–4 hours and features 8–10 hands-on science activities, such as homemade lava bombs, elephant toothpaste, and other engaging elementary-level experiments, designed to spark curiosity, strengthen problem-solving skills, and build a strong foundation in STEM learning.

Two examples of recent workshop activities illustrate the depth and impact of our approach:

- **Cool by Design (Environmental Science & Engineering):** Participants designed and tested insulated water bottle carriers to learn about heat transfer, insulation, and sustainable material choices.
- **Sprouting Science (Agriculture & Sustainability):** Girls explored plant biology and sustainable farming by planting bean seeds to observe at home, supported by donated native seed packets from a local organic nursery.

Grant fund will be used for:

- Subsidize financial assistance for girls and female-identifying youth so that at least 40–50% of participating families can attend workshops at low or no cost. Grant funds will help cover the difference between the \$5–\$30 registration fee (based on financial need) and the actual workshop cost of about \$142 per participant.
- Reduce out-of-pocket costs for families by covering core workshop expenses such as basic materials and classroom supplies, allowing the program to remain affordable and accessible to girls from underrepresented and underserved communities.
- Offset essential program delivery costs (e.g., low-cost materials, printing curriculum handouts, and simple logistical expenses) so that more of the organization's annual budget can be directed toward financial aid and outreach.
- Strengthen volunteer participation by providing minimal training resources (e.g., training materials or background checks) to support high-quality workshop facilitation, thereby increasing capacity to serve more girls at no additional cost.

Workshops will be held in community centers, schools, and libraries across Oregon, Washington, and New Jersey, using accessible spaces that encourage family participation.

Sustainability of Equipment and Facilities

We purchase primarily consumable, take-home materials. Durable, reusable tools (e.g., scissors, measuring tools) will be retained and used across future workshops, ensuring that investment yields long-term value.

Personnel and Qualifications

Our programming is led by an all-women board and staff team with extensive professional experience in STEM fields, ensuring strong mentorship and commitment to equitable programming.

Board Members and Affiliations

- Sarah Foster: Founder & Executive Director, M.S. Biomedical Engineering
- Courtney Davis: Structural Engineer, Dowl
- Jamie Gibbs: Chief of Staff, Nike Global Merchandise Planning
- Aurae Beidler: Assistant Professor, Oregon State University
- Cami Walker: Principal, Bloom Advisory LLC
- Rachelle Cheung: Product Manager, Microsoft
- Gianna Giacolona: Undergraduate Student, Rutgers University

Our workshop volunteers will be recruited through partnerships with local tech, engineering, and research organizations, as well as nearby high schools and colleges. We welcome volunteers of all races, genders, ages, and backgrounds. Building a diverse volunteer team helps create an inclusive environment where participants can see themselves reflected and feel supported.

Target Demographic Population

This project will serve girls in grades 3–5 from underrepresented, low-income, and historically marginalized communities across Oregon, Washington, and New Jersey, with a focus on nine priority counties. We anticipate engaging approximately 150 girls and their caregivers, achieving at least a 25% increase in participant diversity, and providing financial assistance to at least 40% of families.

Workshops were delivered across the following counties:

- Oregon: Multnomah, Washington, Linn-Benton, Marion, Clatsop
- Washington: King, Clark
- New Jersey: Essex, Monmouth

Collaborative Support and Partner Qualifications

We leverage strong partnerships to extend impact:

- Oregon Museum of Science and Industry (OMSI): Provides occasional access to lab materials and professional learning resources.
- Local school districts and community centers: Host workshop spaces and help identify eligible participants.

Project Time Frame

Project Duration: May 2026 – June 2027

Month	Milestone
March–April 2026	Pre-launch preparation during review period (no spending): Begin volunteer outreach, confirm tentative host sites, finalize curriculum drafts, and logistics plans.
May–June 2026	Project launch after funding distribution: Recruit & onboard volunteers, purchase STEM kits and materials, finalize curriculum and workshop materials, and confirm host sites and dates.
July–August 2026	Workshop 1 (delivery + pre/post surveys).
September–October 2026	Workshop 2 (delivery + pre/post surveys).
November 2026 – February 2027	Collect, analyze, and summarize participant data from both workshops, conduct follow-up surveys, review volunteer feedback.
March–June 2027	Final evaluation and reporting: Complete impact report, sustainability plan, next-year workshop calendar, and submit all grant closeout documentation.

EVALUATION

We will implement a structured evaluation process aligned with the project’s goals, primarily focused on increasing confidence, curiosity, and STEM engagement among girls from underrepresented communities.

Objectives and Activities to Be Measured

We will evaluate the following project objectives and core activities:

- Deliver 10–12 hands-on each STEM workshops
- Serve approximately 150 girls and their caregivers.
- Achieve a 25% increase in participant diversity.
- Provide full or partial financial assistance to at least 40% of families.
- Strengthen caregiver involvement in STEM learning.

Quantitative Metrics	Qualitative Metrics
Number of workshops delivered and participants served	Pre/post surveys measuring girls’ confidence and interest in STEM (Likert scale)
Percentage of minorities and underrepresented participants	Caregiver surveys measuring comfort supporting STEM learning at home
Number/percentage of families receiving financial assistance	Open-ended participant feedback on experience and learning gains

Return on Human Investment (RHI)

We will track increases in STEM confidence among girls, caregiver capacity to support STEM learning, and the number of low-income families accessing programs due to financial assistance through surveys and interviews.

Evaluation Timeline

- Staff will collect demographic information and scholarship documentation through the registration form.
- Trained volunteers will track attendance and participation during each workshop. Immediately after each session, participants and caregivers will complete post-surveys and evaluations via paper forms or QR-code surveys.
- The Program Coordinator will compile and analyze all data, preparing a final dataset for use in the grant’s end-of-cycle report and the organization’s annual impact report.

Reporting and Dissemination

Evaluation results will be:

- Compiled into a Final Evaluation Report submitted to the STEM Action Grant Committee
- Shared with internal leadership to refine programming
- Used to inform future grant proposals and strategic planning
- Summarized on our annual community impact report and website

Continuous Improvement Plan

For the next program cycle, we will introduce more consistent STEM interest scales, develop baseline-to-follow-up tracking for return participants, and expand demographic reporting to better identify participation gaps.

FUTURE FUNDING

The programs supported by this grant are designed for long-term sustainability and will continue beyond the funding period by continuing offering community workshops and leverage trained volunteers, as a lasting human capital investment, to expand delivery capacity at low cost.

Following the grant cycle, STEM Like a Girl will sustain programming through a diversified funding model:

- **Corporate Sponsorships and Grants:** We will continue cultivating partnerships with organizations such as Ameresco, Capital Group, and regional funders like the Oregon Community Foundation. We also plan to reapply for the STEM Action Grant, which provides up to three consecutive years of support and subsequently makes us eligible for the one-time \$10,000 Capacity-Building Grant.
- **Program Revenue:** Continuing the sliding-scale participant fees (\$5–\$30 per family) to generate modest, stable revenue.
- **In-Kind Support and Volunteers:** Utilizing our lasting investment in trained volunteers and low-cost workshop space provided by partners.

By increasing operational efficiency and leveraging community resources, we have been able to expand activities while relying on less direct financial assistance. The outcomes—greater confidence and engagement—are designed for long-term impact, leading to growth in repeat participation and expansion into new geographic areas in the medium term.

Continuation of Grant Project Outcomes

The outcomes of this project, greater confidence, curiosity, and engagement in STEM among girls and their caregivers, are designed for long-term impact.

- **Short-Term (Year 1):** Participants demonstrate increased STEM confidence, curiosity, and family engagement as measured by pre- and post-surveys.
- **Medium-Term (Years 2–3):**
 - Growth in repeat participation and returned volunteers.
 - Expansion into new geographic areas through community partnerships.
 - Strengthening site-level volunteer teams.
- **Long-Term (Years 3+):** STEM Like a Girl will expand long-term tracking to understand how early STEM exposure influences future educational pathways.

BUDGET/NARRATIVE

Category	Description	Price	Quantity	Amount (\$)
Workshop Supplies & Materials	Purchase of consumable materials and specialized components for up to 70 participants (e.g., simple circuits, building materials, non-toxic chemicals).	\$21.42	70 participants	1500
Financial Assistance	Full or partial scholarships for up to 70 girls/families (40% target). Covers the gap between true program cost (\$142) and sliding-scale registration (\$5–\$30), ensuring equitable access.	\$25	60 scholarships	1500
Volunteer Recruitment & Training	Training materials, background checks, and modest recognition for 20–36 new volunteer mentors/facilitators. Expands long-term delivery capacity at low cost.	\$13.88	36 volunteers	500
Program Coordination & Evaluation	Staff hours for scheduling, communication, curriculum prep, workshop logistics, and evaluation of pre/post participant and caregiver surveys.	\$15	100 hours	1500
Total				5000

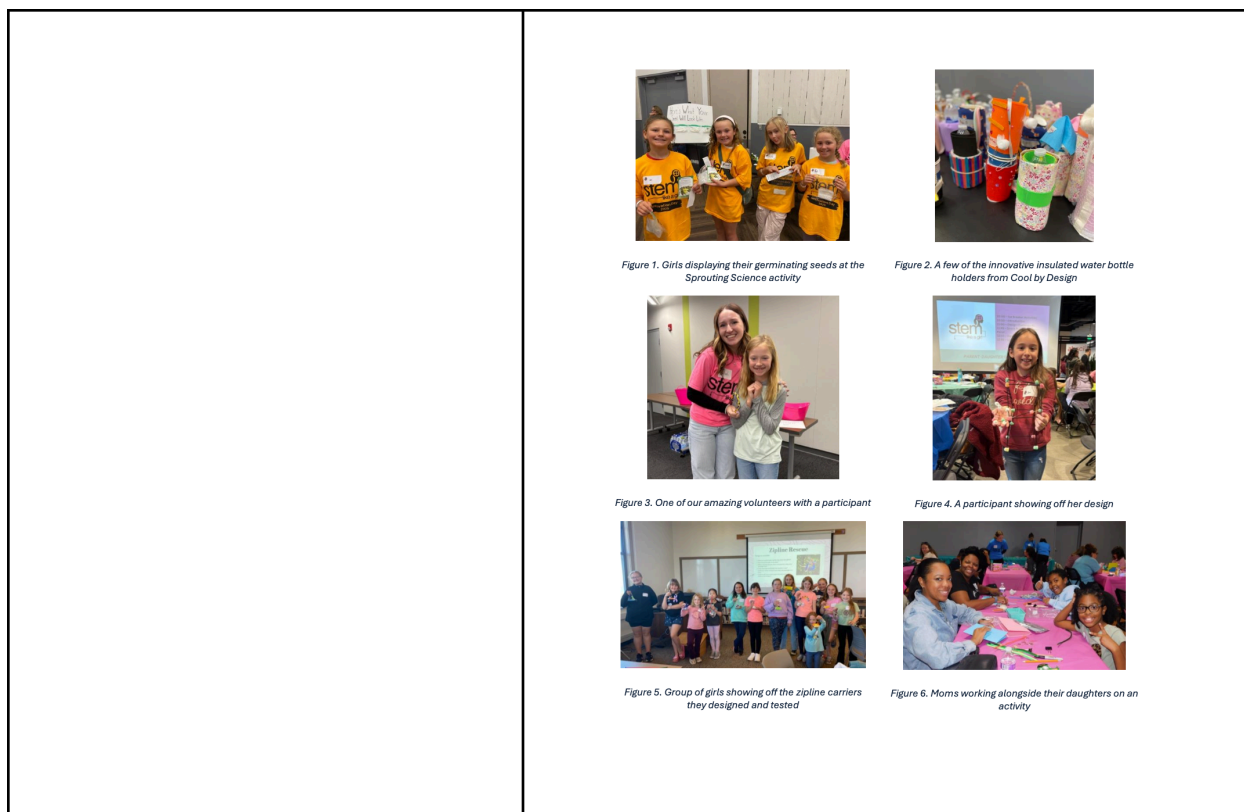
STEM Like a Girl contributes the following matching resources:

- STEM Like a Girl contributes an estimated total value of \$7,000–\$9,000 in matching resources, which is well over 100% of the requested amount. This contribution includes:
 - In-Kind Staff Time: Curriculum development and on-site delivery provided by the Founder and Board.
 - Volunteer Labor: Over 500 hours of donated time from trained volunteer mentors.
 - In-Kind Facility Use: Free or low-cost workshop space provided by partner schools and community centers
 - Earned Revenue: Registration revenue from families not receiving financial assistance.

APPENDIX

Available documents to support the project.

Document	Description
501(c)(3) Certification Letter	Official letter from the Internal Revenue Service (IRS) confirming STEM Like a Girl's tax-exempt, non-profit status.
Letter of Supports	One of the letters of support comes from the Director of Community Engagement for the Technology Association of Oregon and the others are from a mother and her two daughters. These letters confirm the need for our program, attest to the quality of our delivery, and outline specific, current collaboration agreements (e.g., facility usage, volunteer recruitment, or material donation). Action: Ensure the letters are recently dated (within the last 6 months) and clearly reference the scope of this particular project expansion.
Annual Report (Most Recent: 2025)	Provides a comprehensive overview of the organization's financial health, program achievements, previous year participant demographics, and strategic goals.
Budget Report	Covering the period from October 1, 2024 to September 30, 2025
Program Perspective/Curriculum Outline	A short (1-2 page) overview detailing the research-based pedagogy and learning goals for the workshops, including examples of past activities and alignment with educational standards.
Testimonials & Visuals	A portfolio of quotes, high-resolution, permission-granted images, and possibly short narratives from past participants and volunteers, visually reinforcing the positive impact described in the proposal.



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